DAILY TRAIL · METRICS PLAN · NOVEMBER 2026

What we measure in Daily Trail, and what we do about it

Daily Trail is a daily treasure hunt for people who love puzzles. Everyone gets the same hunt each day: a riddle, a cipher and a pattern, whose answers spell a three-letter treasure word. The product works if players come back and finish the hunt week after week.

No real player data yet.

The game is live and already records 13 kinds of player action, such as opening the game, using a hint, finishing and sharing. Those records stay on each player's device for now, so this document is the plan, and nothing in it is a result.

THE BRIEF, AND WHERE EACH PART IS ANSWERED

"Jot down the metrics to be tracked on the game to share inferences to the management and to track what works and doesn't to improve the product."

The metrics to track	The main number and six questions
What works and what doesn't, and how we improve the product	Working, not working, what we change
How inferences reach management	The Monday update
Where the targets come from	How the targets are set
How far to trust the numbers	Read with care

THE MAIN NUMBER

Weekly active solvers

The number of different players who finish at least one hunt in a week, Monday to Sunday. A visit only shows that someone opened the page, while a finished hunt shows they got what they came for. It also tolerates a missed day, so it tracks a habit rather than a perfect streak.

How the targets are set

Each target comes from one of two places, and we do not use numbers from anywhere else.

Published data, for retention

Retention is the one area with credible public figures, so its targets come from the benchmarks below.

Our own first two weeks, for everything else

We found no reliable public figures for start rate, finish rate, hint use, streaks or sharing, so we measure a baseline and set targets from it.

The baseline is the first 14 days of real data, with at least 200 players in the group being measured. Fourteen days shows each of the seven daily hunts twice. A change only counts if it is bigger than the margin of error: about 7 percentage points either way with 200 players, and about 5 with 400.

Published retention benchmarks

Retention	Typical puzzle game	Top quarter of all mobile games (our target)	Top 50 puzzle games
D1	About 20%	**About 27%**	About 43%
D7	About 4.5%	**About 7.5%**	About 25%
D30	About 1.2%	**About 2.8%**	About 16%

Choosing the top quarter is our decision: the game is built for people who already love puzzles, so it should beat a typical game. The top 50 are the most successful puzzle games, so we show them for reference and do not expect to match them.

These figures come from installed mobile apps.
Daily Trail is a website with no install and no notifications. No daily web puzzle publishes its retention, and The New York Times does not release player numbers for its games.

The first two columns use day 28.
That is the closest published figure to day 30. The top 50 figures use day 30.

The targets can change.
We reset them after the baseline if our real audience behaves differently.

SOURCES, CHECKED NOVEMBER 2026, FIGURES ROUNDED

Typical puzzle game and top quarter: GameAnalytics, 2025 Mobile Gaming Benchmarks. 2024 data, 11,600 iOS and Android games. Some top quarter figures are read from charts. gameanalytics.com/reports/2025-mobile-gaming-benchmarks

Top 50 puzzle games: Gamigion, using AppMagic data. Medians. The article does not state its year or how the games were selected. gamigion.com/benchmarks-from-the-top-50-mobile-puzzle-games

Streak freeze: Duolingo, January 2022. Reported by the company and not independently verified. blog.duolingo.com/how-duolingo-streak-builds-habit

Six questions we answer each week

Each question lists its metrics and targets, then what working and not working look like, and the change we make when it is not working. "Clearly" means by more than the margin of error for the group size.

Do players come back?

D1 retention

New players who open the game exactly 1 day after their first visit

About 27%

D7 retention

The same, exactly 7 days later

About 7.5%

D30 retention

The same, exactly 30 days later

About 2.8%

Solves per active player

Finished hunts per week, divided by weekly active solvers

Set from baseline

Working

- Retention at or above the top quarter of mobile games, and weekly active solvers growing week on week.

Not working, and what we change

- D1 under 20%, the typical puzzle game: the first play gave no reason to return.
  Show tomorrow's hunt and the streak more clearly on the result screen, then test a reminder.
- D1 of 20% or more but D7 under 4.5%: the novelty wears off, or players notice the hunts repeat.
  Add new hunts first, then look at weekly goals.
- D7 of 4.5% or more but D30 under 1.2%: regular players run out of reasons to stay.
  Build longer-term goals, such as crews or a monthly story map.

Do they finish?

Start rate

Players who press Start, out of players who open the game and have not finished today

Set from baseline

Finish rate

Players who finish, out of players who press Start that day

Set from baseline

Stage drop-off

Players who see a stage but do not solve it, for the riddle, cipher and pattern separately

Compare the three stages

Working

- Start and finish rates hold or rise against their baseline, and no stage loses clearly more players than the other two.

Not working, and what we change

- Start rate falls clearly below its baseline: the intro does not make players want to begin.
  Shorten the intro and show the first puzzle sooner.
- Finish rate falls clearly below its baseline, and one stage loses clearly more players than the other two: that stage is the blocker.
  Fix that stage type before writing more content.
- The cipher loses clearly more players than the riddle and the pattern: its rules are too hard to follow.
  Simplify the "How this works" panel and strengthen the first hint.

Are the puzzles right?

Clean solve rate
Stage solves with no hint, by puzzle and stage
Compare within type

Answer shown rate
Stage solves where the last hint showed the answer
Compare within type

Wrong answers per stage
Wrong answers divided by players who saw the stage
Compare within type

Median solve time
The middle time for each stage and for the whole hunt
Compare within type

Working

- No puzzle stands out as clearly easier or harder than others of its type.

Not working, and what we change

- A puzzle's clean solve rate is clearly above others of its type: too easy for this audience.
  Make that puzzle harder.
- A puzzle's answer shown rate is clearly above others of its type: too hard or unfair.
  Rewrite the puzzle or its hints, and retire it if that fails.
- A puzzle has clearly more wrong answers than others of its type, but no more hints: we may be rejecting fair answers, or players may not see the hints.
  Review accepted answers and make the hint button easier to see.
- Finish rate rises while D7 falls: the hunts may have become too easy to feel rewarding.
  Roll back the easier content and check the difficulty mix.

Does the streak build a habit?

Streak of 3 or more

Weekly active solvers holding a streak of at least 3 days

Set from baseline

Freeze return rate

Players who finish a hunt within 2 days after the weekly freeze covered a missed day

Compare with next row

Return after a broken streak

Players who finish a hunt on the day their streak resets, or the day after

Compare with row above

Working

- Players saved by a freeze return clearly more often than players whose streak breaks.

Not working, and what we change

- The two groups return at about the same rate: the freeze may not be worth its complexity.
  Simplify the streak rules.
- Freezes clearly help: the habit is fragile around missed days. Duolingo saw a relative 0.38% rise in daily active learners from allowing a second freeze, so expect a small effect.
  Test a second freeze, or a way to earn one.

Do players spread it?

Share rate

Players who use Share or Copy on their result, out of players who finish that day. The result card is spoiler free: coloured squares, time and streak.

Set from baseline

Working

- Share rate holds or rises against its baseline, and new players grow alongside shares.

Not working, and what we change

- Share rate falls clearly below its baseline: the result is not worth showing off.
  Test a new share card and a head-to-head challenge link.
- Share rate holds but new players stay flat: shares do not bring people in.
  Track arrivals from shared links, then change the message or the link.

Do players want more than one hunt a day?

Practice rate

Players who open a practice hunt, out of players who finish that day. Practice hunts are separate from the daily hunts and never count.

Set from baseline

Replay rate

Players who look back at a past hunt's answers or play it again, out of players who open the game that day

Set from baseline

What a rising rate tells us

- Practice rate rises clearly above its baseline: more players want more than one hunt a day.
- Replay rate rises clearly above its baseline: players value their record of past hunts.

What we change

- **Add more practice hunts or a weekly bonus hunt, and check it does not lower next-day returns.**
- **Make past hunts more prominent, for example with a monthly map of finished hunts.**

How inferences reach management

Each Monday, management gets one short update with four parts:

1. Weekly active solvers, against last week.
2. D1 and D7 for the newest group of players with enough time to measure.
3. The hardest and the easiest puzzle of the week.
4. One thing we learned, one decision we made, and what we will check next week.

TEMPLATE, TO BE FILLED IN ONCE DATA EXISTS

Weekly active solvers: _____, against _____ last week.

D1: _____. D7: _____.

Hardest puzzle: _____. Easiest: _____.

Learned: _____. Decided: _____. Checking next week: _____.

Read with care

Check the group size first.
With 100 players, a measured 45% could really be anywhere from about 35% to 55%. Small differences mean little until each group has several hundred players.

A player means one browser on one device.
Someone on a second device, in a private window or after clearing browser data counts as a new player. This inflates new players and lowers retention a little.

Streak numbers miss players who leave for good.
A streak change is only recorded when a player opens the game, so read streak numbers alongside retention.

Freeze comparisons do not prove cause.
Players who keep a streak long enough to use a freeze may already be more committed. An A/B test is needed before we credit the freeze.

Each puzzle always falls on the same weekday.
There are seven daily hunts on a seven-day cycle, so a hard puzzle and a slow weekday look the same.

Hunts repeat after seven days.
Returning players may remember the answers, so we judge puzzle difficulty on first plays only.

A share is an intent, not a delivery.
Copy shows that the text was copied, not that it was posted.

The leaderboard is sample data.
Only the player's own row is real, so it tells us nothing about competition yet.

What we need before the numbers are real

Collect events in one place

Today each player's events stay on their own device, so we cannot see any numbers across players.

Fix: send events to an analytics service. This comes first, because nothing else on this page works without it.

Tag shared links

We cannot tell whether shares bring in new players.

Fix: add a tag to the link in the share text and record it on a player's first visit.

Keep wrong answers

We cannot tell typos and fair alternative answers from real mistakes.

Fix: record a privacy-safe form of wrong answers and review the most common ones.

Learn why players stop

Leaving the page also happens when someone briefly switches apps, so it only shows where a player was.

Fix: use "started but did not finish that day" as the measure, and add a one-question exit survey if needed.

Play the game: dailytrail.vercel.app

The same plan, with exact event definitions, is kept with the game's code as docs/METRICS.md and reference/events.md.